Ethical Hacking Internship

Week 1 Manual

Lab Setup and Target Enumeration

Program: Ethical Hacking Internship

Week	Week 1
Focus	Lab Setup and Target Enumeration

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1. Weekly overview

Week 1 introduces the operational foundation of the Ethical Hacking Internship. Before students can scan, test, or exploit anything responsibly, they need a stable virtual lab, correct network visibility, and a disciplined approach to identifying hosts, ports, and services. This week is therefore focused on environment preparation and target enumeration rather than advanced attack execution.

2. Class focus

Students are expected to learn how to prepare a working lab and enumerate a target correctly without wasting time on unnecessary theory. The emphasis is on practical readiness: building the lab, confirming connectivity, discovering live hosts, and identifying exposed services.

3. Topics covered

Topic Area	Details
Lab preparation	Lab setup with Kali Linux and vulnerable targets
Networking	Host-only or internal virtual networking
Target identification	Identifying lab target IP addresses
Reconnaissance	Active reconnaissance within lab scope
Scanning	Nmap basics
Enumeration	Service and port enumeration

4. Weekly task requirements

The weekly task is not one single action. It is a chain of proof that shows the student can build a lab, verify communication, discover systems, enumerate services, and explain what the findings mean. Each requirement below should be completed in order because later steps depend on earlier ones.

Requirement	What the student must do	Submission evidence
Build the lab environment	Install a Kali Linux virtual machine and at least one vulnerable machine such as Metasploitable 1, 2, or 3 using VMware Workstation or VirtualBox. Configure both systems so they can communicate with each other inside the lab.	Screenshots of the Network Interface settings for both virtual machines.
Confirm internal and external connectivity	Use the ping command from each machine to verify that internal communication between the virtual machines works and that external internet access is also functioning where required.	Screenshots showing successful ping results for both internal VM-to-VM communication and external connectivity.
Identify live hosts in the lab	Use arp-scan or Nmap from the Kali Linux machine to identify and list all live hosts available in the virtual lab environment.	Screenshots of the command used and the scan results showing active hosts.
Perform active	Use Nmap from Kali Linux to identify open ports and running	Screenshots of the commands used and the resulting port and

reconnaissance on the target	services on the target machine in the lab environment.	service enumeration output.
Write a short target note	Write a concise note about the target system you enumerated, such as Metasploitable 2. Mention the open ports discovered, the services running on them, what the results reveal about system configuration, and what possible vulnerabilities or attack surfaces these findings may indicate.	A short written analysis based on the Nmap findings.

5. Submission evidence checklist

Students should not submit random screenshots. The submission should clearly prove that the lab was built correctly and that enumeration was carried out in a logical sequence. The checklist below converts the weekly task into a practical submission standard.

#	Required submission item
1	Kali Linux virtual machine installed and functional
2	At least one vulnerable target machine installed and functional
3	Network interface settings captured for both virtual machines
4	Successful internal connectivity verified with ping
5	Successful external connectivity verified where required
6	Live-host discovery completed with arp-scan or Nmap
7	Port and service enumeration completed with Nmap
8	Short written note included explaining findings and possible attack surface